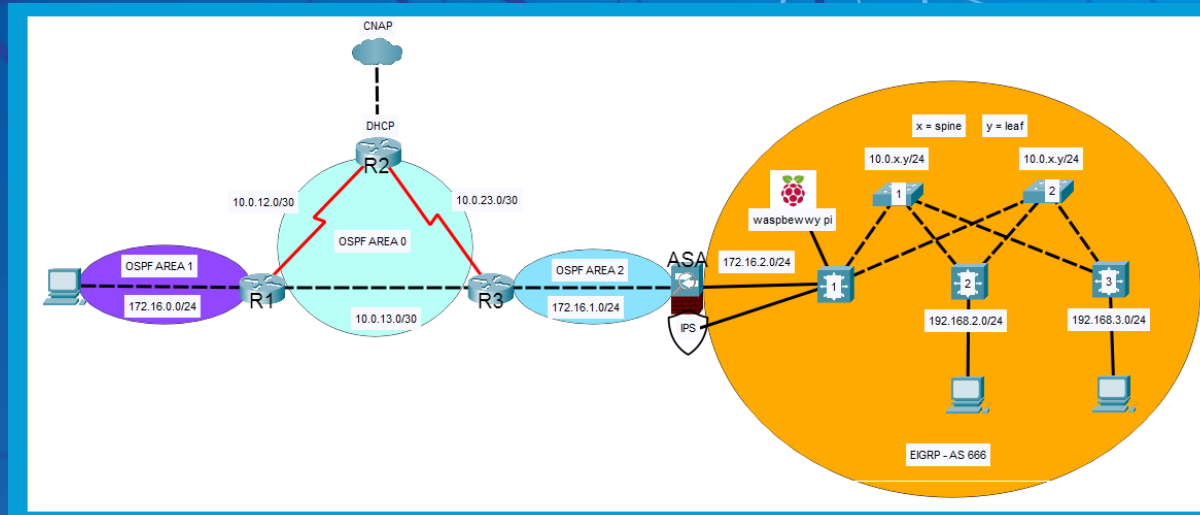


CASE 5

IPS

2023-12-10



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Om denna Rapport

Skynet Industriars ökar i otrolig fart tack vare deras T-101 modeller. Dem behöver hjälp med att sätta upp en IPS för ASA 5510, IPS modulen som användes är "SSM-AIP10-k9".

För att testa att IPS modulen fungerar testades en "directory traversal attack" mot webbservern på insidan.

Ytligare säkerhetsmekanismer har implementerats för att säkerställa ytligare säkerhet.

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1 Inledning / Topologi

Denna topologi lägger en stor vikt på East-West trafik då de anställda ska ha snabbt uppkoppling mot de intärna tjänsterna men inte mot Internet. För att göra det så snabbt som möjligt för inside hosts har en spine-leaf arkitektur implementerats på insidenätverket. För att testa attacker jämt mot insidenätverket har en host placerats i AREA 1. En EDGE router placerades även mot CNAP nätet för att kunna komma åt Internet (se *figur 1*). EDGE routern utför PAT med adressen den fått via DHCP från CNAP. Den översätter alla adresser.

```
ip nat inside source list NAT interface GigabitEthernet0/0/1 overload
```

figur 1 - NAT mot CNAP

1.1 Routing

Både OSPF och EIGRP används i denna topologi. Som topologi-bilden hänvisar finns 3 olika areor, ASAn redistribuerar in EIGRP processen in i OSPF processen. EDGE routern delar ut sin default route till övriga routrar. På insidenätverket (den orangea cirkeln) körs EIGRP, ASAn distribuerar in en default route till AS 666. *Figur 2 & 3* visar routingtabellerna för respektive ASAn och R2. Som man kan se i *figur 2*, ASAns tabell finns både routes från samma- och olika areor med OSPF samt EIGRP routes från insidenätverket. I övriga enheters tabeller syns EIGRP routes som externa, se *figur 3* för R2's exempel. *Figur 4* visar hur *Leaf-1's* routingtabell ser ut.

```
Gateway of last resort is 172.16.1.1 to network 0.0.0.0

O IA 172.16.0.0 255.255.255.0 [110/12] via 172.16.1.1, 0:06:47, OUTSIDE
C 172.16.1.0 255.255.255.0 is directly connected, OUTSIDE
C 172.16.2.0 255.255.255.0 is directly connected, inside
O IA 10.0.12.0 255.255.255.252 [110/507] via 172.16.1.1, 0:44:55, OUTSIDE
O IA 10.0.13.0 255.255.255.252 [110/11] via 172.16.1.1, 0:44:55, OUTSIDE
D 10.0.2.0 255.255.255.0 [90/28416] via 172.16.2.2, 0:32:13, inside
D 10.0.1.0 255.255.255.0 [90/28416] via 172.16.2.2, 0:32:13, inside
O IA 10.0.23.0 255.255.255.252 [110/510] via 172.16.1.1, 0:44:55, OUTSIDE
D 192.168.2.0 255.255.255.0 [90/28672] via 172.16.2.2, 0:14:53, inside
D 192.168.3.0 255.255.255.0 [90/28672] via 172.16.2.2, 0:17:17, inside
Gateway of last resort is 172.16.2.1 to network 0.0.0.0
```

figur 2 – ASAn's routingtabell

```
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PF
Gateway of last resort is 193.10.203.129 to network 0.0.0.0

S* 0.0.0.0/0 [254/0] via 193.10.203.129
10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
O E1 10.0.0.0/22 [110/622] via 10.0.23.1, 01:42:44, Serial0/1/1
C 10.0.12.0/30 is directly connected, Serial0/1/0
L 10.0.12.2/32 is directly connected, Serial0/1/0
O 10.0.13.0/30 [110/501] via 10.0.23.1, 03:32:34, Serial0/1/1
C 10.0.23.0/30 is directly connected, Serial0/1/1
L 10.0.23.2/32 is directly connected, Serial0/1/1
172.16.0.0/24 is subnetted, 3 subnets
O IA 172.16.0.0 [110/501] via 10.0.12.1, 00:08:59, Serial0/1/0
O IA 172.16.1.0 [110/501] via 10.0.23.1, 01:48:33, Serial0/1/1
O E1 172.16.2.0 [110/622] via 10.0.23.1, 01:42:50, Serial0/1/1
O E1 192.168.2.0/23 [110/622] via 10.0.23.1, 01:42:44, Serial0/1/1
193.10.203.0/24 is variably subnetted, 2 subnets, 2 masks
C 193.10.203.128/25 is directly connected, GigabitEthernet0/0/1
```

figur 3 – R2's routingtabell

```
D*EX 0.0.0.0/0 [170/28416] via 172.16.2.1, 00:47:56, GigabitEthernet1/0/24
10.0.0.0/8 is variably subnetted, 5 subnets, 3 masks
D 10.0.0.0/22 is a summary, 00:48:02, Null0
C 10.0.1.0/24 is directly connected, GigabitEthernet1/0/1
L 10.0.1.1/32 is directly connected, GigabitEthernet1/0/1
C 10.0.2.0/24 is directly connected, GigabitEthernet1/0/2
L 10.0.2.1/32 is directly connected, GigabitEthernet1/0/2
172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.16.2.0/24 is directly connected, GigabitEthernet1/0/24
L 172.16.2.2/32 is directly connected, GigabitEthernet1/0/24
D 192.168.2.0/23 is a summary, 00:48:02, Null0
D 192.168.2.0/24 [90/3072] via 10.0.2.2, 00:35:50, GigabitEthernet1/0/2
[90/3072] via 10.0.1.2, 00:35:50, GigabitEthernet1/0/1
D 192.168.3.0/24 [90/3072] via 10.0.2.3, 00:35:52, GigabitEthernet1/0/2
[90/3072] via 10.0.1.3, 00:35:52, GigabitEthernet1/0/1
Leaf-1#
```

Figur 4 - Leaf-1's routingtabell

1.2 Access-layer

Vi ville skydda access-lagret lite extra och underlätta de anställda med dynamisk adressering. Portar som inte användes blockerades och lades till i ett annat VLAN (se *figur 5 för demonstration på Leaf-1*). DHCP användes även, se *figur 6* som visar poolerna för DHCP. R3 agerar som en DHCP server för alla nätverk med hostar. Interfacerna behövde en *helper-adress* då servern låg på ett annat nätverk, *figur 7* visar hur det kan se ut. *Figur 8 & 9* visar hur utdelning av en adress kan se ut för en respektive host och en router. Vi valde även att hostarna skulle ha access mot internet. *Figur 10* visar när en host kan pinga youtube & google.

```
Leaf-1(config)#do sh int status

Port      Name      Status      Vlan      Duplex      Speed Type
Gi1/0/1   Link to Spine-1   connected   routed    a-full      a-1000  10/100/1000BaseTX
Gi1/0/2   Link to Spine-2   connected   routed    a-full      a-1000  10/100/1000BaseTX
Gi1/0/3   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/4   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/5   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/6   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/7   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/8   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/9   NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/10  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/11  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX

Port      Name      Status      Vlan      Duplex      Speed Type
Gi1/0/12  in use     monitoring   routed    a-full      a-1000  10/100/1000BaseTX
Gi1/0/13  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/14  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/15  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/16  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/17  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/18  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/19  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/20  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/21  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/22  NOT IN USE        disabled    50        auto        auto    10/100/1000BaseTX
Gi1/0/23  NOT IN USE        notconnect  routed    auto        auto    10/100/1000BaseTX

Port      Name      Status      Vlan      Duplex      Speed Type
Gi1/0/24  link to ASA   connected   routed    a-full      a-100   10/100/1000BaseTX
Gi1/1/1   NOT IN USE        disabled    50        auto        auto    unknown
Gi1/1/2   NOT IN USE        disabled    50        auto        auto    unknown
Gi1/1/3   NOT IN USE        disabled    50        auto        auto    unknown
Gi1/1/4   NOT IN USE        disabled    50        auto        auto    unknown
```

figur 5 – Portar på Leaf-1

```
R3(config)#do sh run | s dhcp
ip dhcp excluded-address 192.168.2.1 192.168.2.10
ip dhcp excluded-address 192.168.3.1 192.168.3.10
ip dhcp excluded-address 172.16.0.1 172.16.0.10
ip dhcp pool AREA_1
network 172.16.0.0 255.255.255.0
default-router 172.16.0.1
domain-name cisco.com
dns-server 8.8.8.8
ip dhcp pool LAN2
network 192.168.2.0 255.255.255.0
domain-name cisco.com
dns-server 8.8.8.8
default-router 192.168.2.1
ip dhcp pool LAN3
network 192.168.3.0 255.255.255.0
default-router 192.168.3.1
domain-name cisco.com
dns-server 8.8.8.8
```

figur 6 – DHCP Pool

```
R3(config)#do sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration        Type      State      Interface
                Hardware address/
                User name
172.16.0.11     0118.0373.25f5.0f    Dec 09 2023 03:44 PM    Automatic Active      GigabitEthernet0/0/0
192.168.2.11    0134.17eb.df13.ba    Dec 09 2023 03:14 PM    Automatic Active      GigabitEthernet0/0/1
192.168.3.11    0118.0373.20bc.78    Dec 09 2023 03:15 PM    Automatic Active      GigabitEthernet0/0/1
```

figur 9 – DHCP adress-utdelning, router

```
interface GigabitEthernet0/0/1
ip address 172.16.0.1 255.255.255.0
ip helper-address 10.0.13.2
negotiation auto
```

figur 7 – IP Helper-adress

```
C:\Users\cisco>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=12ms TTL=48
Reply from 8.8.8.8: bytes=32 time=12ms TTL=48

Ping statistics for 8.8.8.8:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 12ms, Average = 12ms
Control-C
^C
C:\Users\cisco>ping youtube.com

Pinging youtube.com [172.217.21.174] with 32 bytes of data:
Reply from 172.217.21.174: bytes=32 time=12ms TTL=106
Reply from 172.217.21.174: bytes=32 time=12ms TTL=106
Reply from 172.217.21.174: bytes=32 time=35ms TTL=106
```

figur 10 – Host kan pinga youtube & google

```
C:\Users\cisco>ipconfig /renew

Windows IP Configuration

Ethernet adapter Ethernet 5:

Connection-specific DNS Suffix . : cisco.com
Link-local IPv6 Address . . . . . : fe80::3ead:d199:a546:b3da%19
IPv4 Address. . . . . : 192.168.2.11
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.2.1
```

figur 8 – DHCP adress-utdelning, host

1.3 SPAN

Då en ytterligare säkerhetsmekanism implementerats har SPAN behövts användas på Leaf-1 switchen. G1/0/24 porten monitoreras (porten mot ASAn) och skickas sedan mot g1/0/12 (porten mot RPIn).

```
Leaf-1(config)#do show monitor session 1
Session 1
-----
Type                : Local Session
Source Ports        :
    Both            : Gi1/0/24
Destination Ports   : Gi1/0/11-12
Encapsulation       : Native
Ingress              : Disabled
```

figur 11 – SPAN

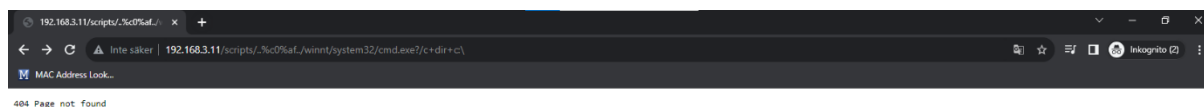
2 IPS

För att visa att IPS:en fungerar så används specialsträngen som Skynet gav till oss som kommer läggas på [URL:en](#) till hemsida på vår http server.

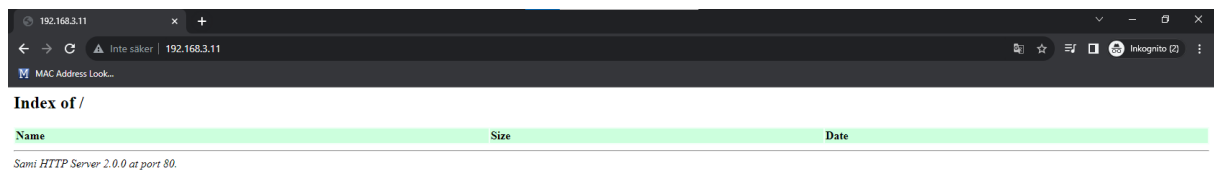
När vi först utförde ett test innan IPS:en hade aktiverats så fick vi svaret som visas i *figur 12*, att sidan inte går att ladda. Själva sidan laddar korrekt om extrasträngen inte används vilket ses i *figur 13*. Detta betyder att http servern fungerar men inte med strängen pålagd.

Efter att IPS:en har aktiverats så blockerar IPS:en återkomst till websidan eftersom den matchar en signatur som finns på IPS:en och i meddelandet som visas i webbläsaren så ses det att anslutningen har blivit återställd vilket ses i *figur 14*

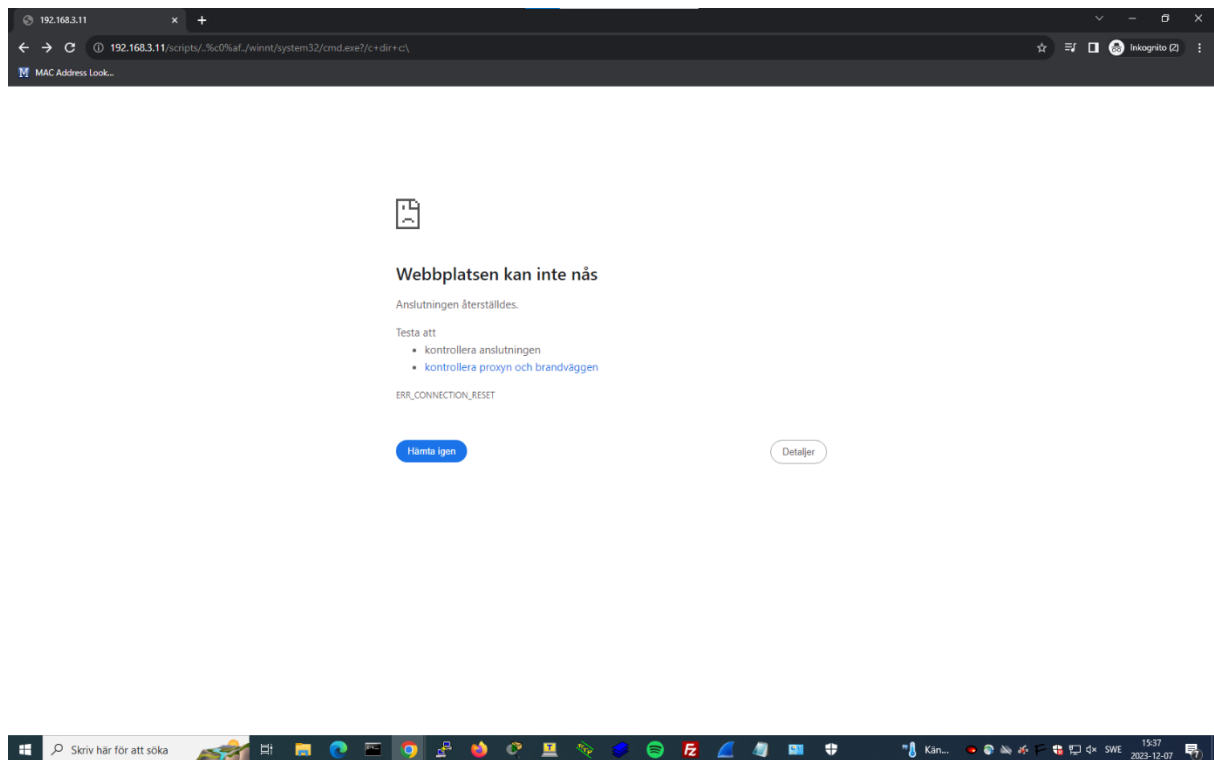
I loggen som visas i *figur 15* så ser man att IPS:en har blockerat paketen som använder sig av extrasträngen. Den skickar en "TCP Reset" till datorn som startade anslutningen för att kapa den och den visar vad för typ av attack det är under "Events" fliken.



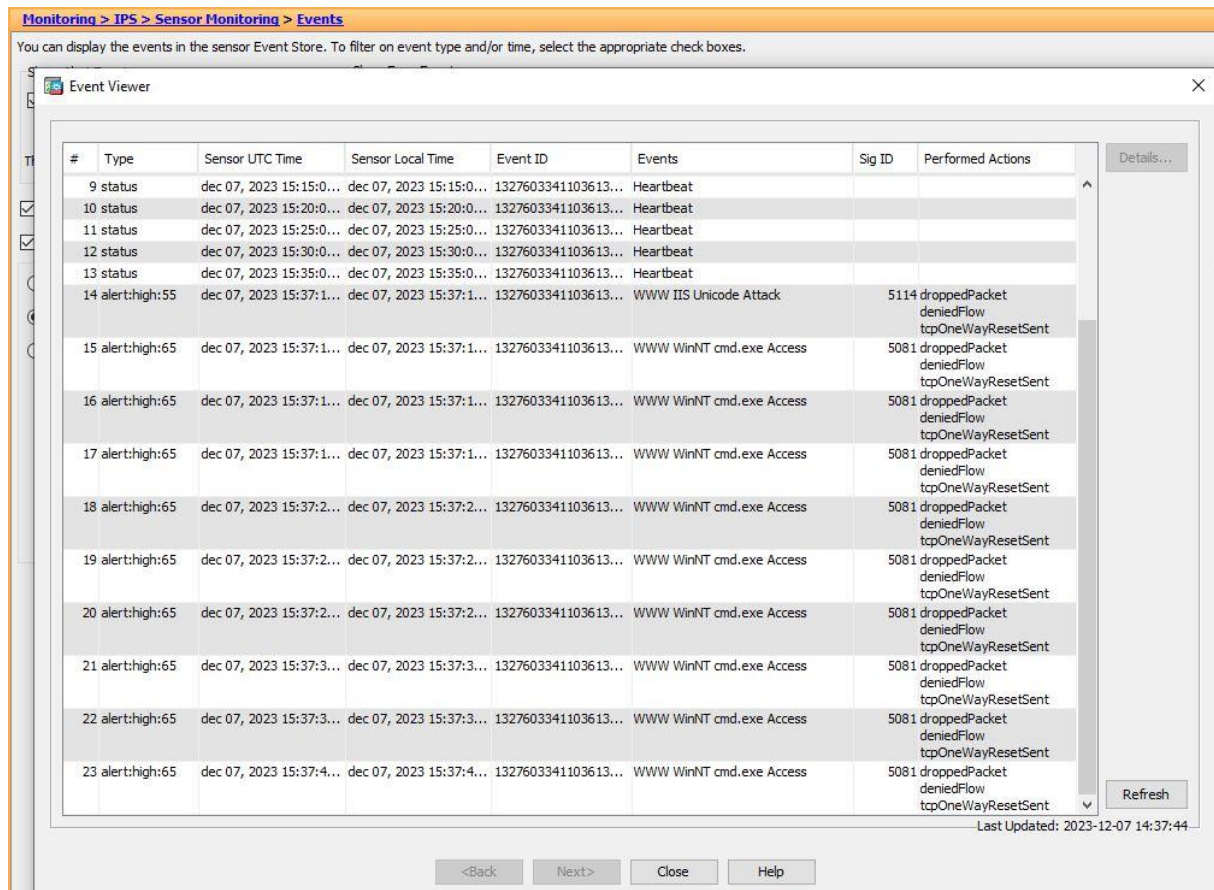
Figur 12: Test utan IPS aktiverat



Figur 13: Test där hemsidan nås



Figur 14 IPS:en blockerar åtkomst till webbsidan



Figur 15: IPS event monitor

3 IPS

I figur 16 skapas en access lista på vår ASA som vi döper till Traffic_For_Ips, sedan så tar vi och skriver permit ip any any vilket betyder att vi tillåter all IP trafik från vilken källa som helst till vilken destination som helst. Efter det så skapar vi en class-map som heter ips_class_map som vi sedan matchar till vår access-lista Traffic_For_Ips

```
ASA(config)# access-list Traffic_For_Ips permit ip any any
ASA(config)# class-map ips_class_map
ASA(config-cmap)# match access-list Traffic_For_Ips
ASA(config-cmap)#
```

Figur 16: Accesslista för IPS

```
The following configuration was entered.

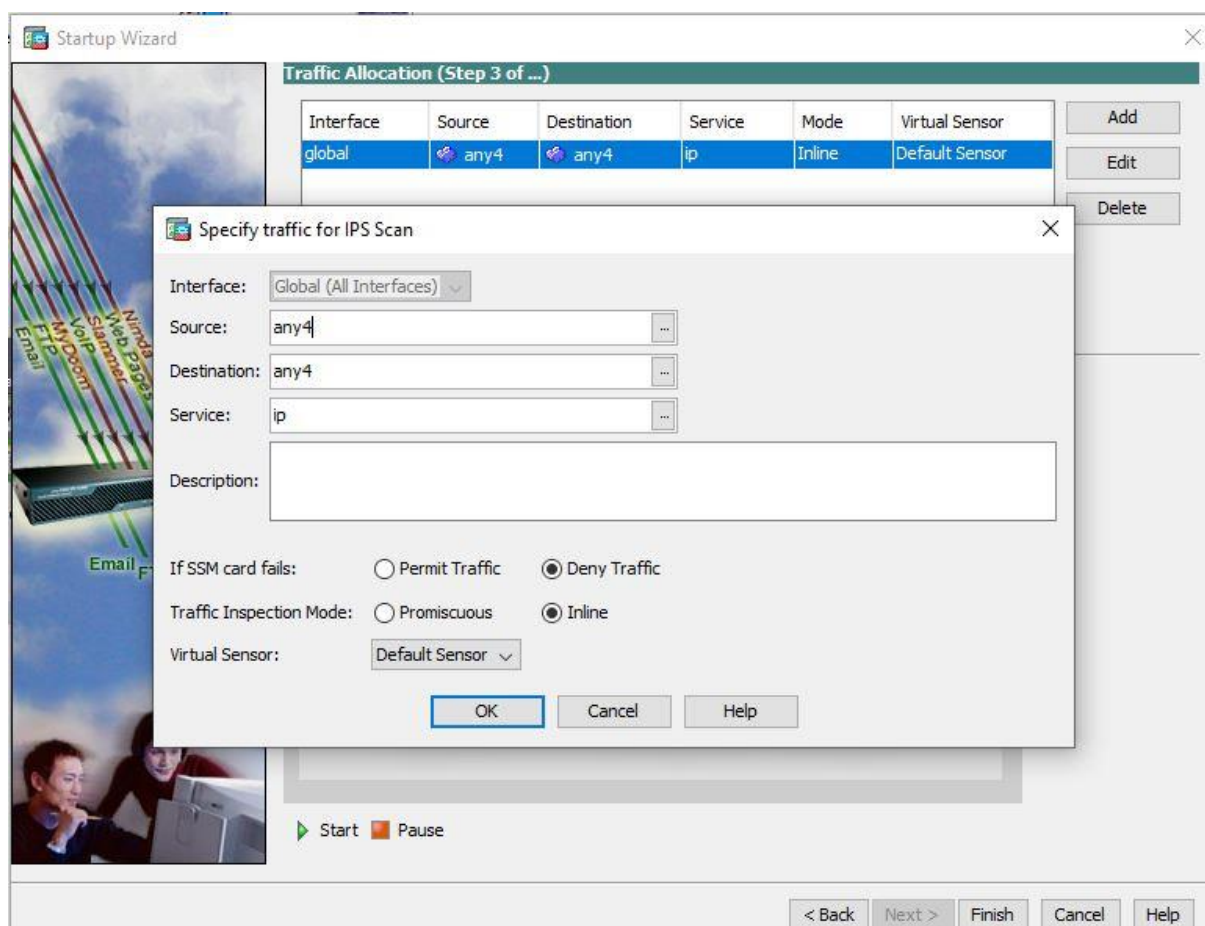
service host
network-settings
host-ip 192.168.69.5/24,192.168.69.1
host-name IPS
telnet-option disabled
ftp-timeout 300
no login-banner-text
dns-primary-server disabled
dns-secondary-server disabled
dns-tertiary-server disabled
http-proxy no-proxy
exit
time-zone-settings
offset 0
standard-time-zone-name UTC
exit
summertime-option recurring
offset +1
summertime-zone-name UTC
start-summertime
month march
week-of-month second
day-of-week sunday
time-of-day 02:00:00
exit
end-summertime
month november
week-of-month first
day-of-week sunday
time-of-day 02:00:00
exit
ntp-option disabled
exit
service global-correlation
network-participation off
exit

[0] Go to the command prompt without saving this config.
[1] Return to setup without saving this config.
[2] Save this configuration and exit setup.
[3] Continue to Advanced setup.

Enter your selection[3]: 2
```

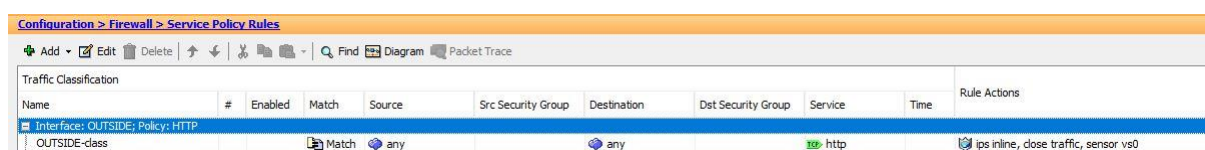
Figur 17: IPS grund konfiguration

I figur 17 ses basic config som görs för vår IPS och IPS står för Intrusion Prevention System vilket betyder att IPS funktion är att övervaka nätverkstrafik för att sedan kunna avgöra vad som är intrångsförsök och vad som är skadlig aktivitet. Detta gör den genom att identifiera trafiken som kommer in på vår IPS. IPS analyserar trafiken i realtid och kan direkt märka om något mönster bryter sig eller upptäcker kända signaturer för attacker. IPS kommer då att agera direkt för att blockera eller hindra dessa hot från att komma in i nätverket.



Figur 18: ACL För IPS i ASDM

I figur 18 ser man hur vi bara specificerar trafiken för vår IPS scan genom att skriva från vilken source den ska gå till vilken destination och med service IP. Det som alltså går från den sourcen till själva destinationen är inline och kommer därför att denas alltså blockas.



Figur 19: IPS service policy

Efter vår konfiguration för IPS så lade vi till en service-policy som ses i figur 19 för att kunna dirigera HTTP trafik till IPS för att sedan ytterligare kunna inspektera samt skydda vår IPS. Service-policyn fungerar som en gateway och riktar all inkommande HTTP trafik till IPS. Där kommer trafiken sedan att analyseras för att upptäcka eventuella intrångsförsök eller skadlig aktivitet. Själva konfigurationen är väldigt skalbar eftersom fler tjänster kan enkelt läggas till vid behov. Detta kommer att öka flexibiliteten och säkerhetsinfrastrukturen och sedan anpassa den efter de specifika kraven och hot som kan uppstå i framtiden.

4 Problem

I dagens uppkopplade samhälle så kommer det nya hot och cyber kriminellas angreppsförsöken uppdateras ständigt för att undvika att bli påtagna. När IPS:n kollades igenom så noterades det att signaturer uppdaterades senaste gången över 13 år sedan. (se figur 20)

Då hoten idag är alltmer komplicerade än vad dem var för 13 år sedan så kommer er IPS ofta att missa attacker och då visa "false negative".

Då ni har haft problem med praktikanter som sovit tidigare under arbetstid och Cisco ASA SSM-AIP 10 bara gett en log att något har hänt så har vi kommit på en lösning för er.

This represents overall sensor health. Sensor health can be shown as Normal, Needs Attention, or Critical. Sensor health is calculated based on the following metrics.

Metric	Status	Current	Yellow ...	Red Thr
Inspection Load	✓ Normal	1%	80%	91%
Missed Packet	✓ Normal	0%	1%	6%
Signature Update	✗ Critical	4886 days	30 days	60 days
License time remaining	✗ Critical	No License	30 days	0 days
Event Retrieval	✗ Critical	Not retrieved	300 sec	600 sec

Figur 20

4.1 Lösning

För att kunna ge er en ökad säkerhet samt göra så att anställda märker av när något har hänt på ert nätverk så sattes ytterligare säkerhets hårdvara in.

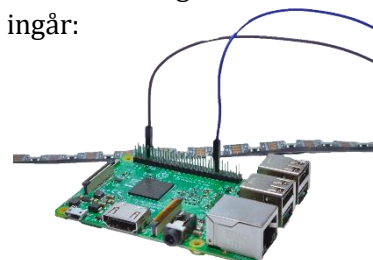
4.1.1 IDS

För att inte bara tänka på säkerhetens skull utan också prestandan så valdes det att sätta in en IDS som tar emot en kopia av trafiken som IPS:n tillåter in i det interna nätverket. IDS:n i frågan är vår BDsmIDS som aktivt kör vår IDS mjukvara "WaTFU".

4.1.2 BDsmIDS funktioner

BDsmIDS är en liten dator som sätta in i nätverket och som tar emot en kopia av nätverks trafiken genom till exempel SPAN. Som alla IDS skickas notiser ifall trafiken den tar emot stämmer överens med dem säkerhetsregler som finns på plats. Allting körs på datorn i sig och får uppdateringar med nya regler när nya hot har analyserats av någon av våra avdelningar har bekräftat och skrivit en ny regel. Några av mjukvara funktionerna som ingår:

- Stor mängd signatur-baserade regler
- Stöd för flera nätverksprotokoll som TCP, UDP, ICMP och IP
- IPV4 whitelisting
- Dynamisk uppdatering av regler
- Detekterar skanningar av nätverket



Det är inte i mjukvaran som gör vår IDS annorlunda från andras utan att den har också dem fysiska funktioner. IDS:n kommer med en led list och funktioner som gör så att den också går att koppla in till externa ljudenheter som spelar upp valfritt ljud vid valda regler. Till skillnad från IPS:n så ger BDsmIDS ut en tydlig signal av att den har identifierat trafik som indikera att någon har eller försöker få tillgång till nätverket.

4.1.3 Varför båda?

En IPS och en IDS jobbar väldigt bra ihop då båda har sina för- och nackdelar där dem kompenserar för varandra. En IPS har en liten påverkan på prestandan på nätverken när det kommer till latency och jitter medan IDS:n inte har någon alls då den bara tar emot en kopia av trafiken. IDS:n stoppar tyvärr inga packet som den ser som farliga vilket en IPS gör men i och med att det var en väldigt lång tidsedan den fick en uppdatering med uppdaterade signaturer så märker den inte av många av dagens hot, det är här vår IDS med regler anpassade för hoten som väntar på en väg in.

4.1.4 Konfiguration:

För att göra så att IDS:n tar emot trafik så användes Span, detta uppfylldes med funktionen "monitor". Kommandona som användes var: *monitor session 1 source interface Gi1/0/24* interfacet G11/0/24 är mot "outside" det andra kommandot som användes för att skicka trafiken till IDS:ns port var: *#monitor session 1 destination interface g1/0/12* med dessa två kommandon skickas en kopia av trafiken som går genom interface Gi1/0/24 till Gi1/0/12 där IDS:n är kopplad.



Glöm inte att sätta på promiscuous mode så den kan ta emot trafiken på IDS:ns nätverkskort med kommandot:

```
# ip link set eth0 promisc on
```

4.1.5 Verifiera Körande

När det är dags att sätta på IDS mjukvaran så körs *#sudo systemctl enable watfu.service* samt *#sudo systemctl start watfu.service*. för att verifiera att det körs så går det att kolla statusen för mjukvaran med *#sudo systemctl status watfu.service*. (se figur 21)

```
bizken@raspberrypi:~ $ sudo systemctl status watfu.service
* watfu.service - WaTFU IDS daemon
   Loaded: loaded (/lib/systemd/system/watfu.service; enabled; preset: enabled)
   Active: active (running) since Thu 2023-12-07 22:53:21 CET; 1h 11 min ago
     Docs: README.md
   Process: 855 ExecStart=/usr/bin/watfu -D --af-packet -c /etc/watfu/watfu.yaml
  Main PID: 896 (watfu-Main)
    Tasks: 10 (limit: 1559)
       CPU: 5.573s
   CGroup: /system.slice/watfu.service
           └─896 /usr/bin/watfu -D --af-packet -c /etc/watfu/watfu.yaml --pidfile
```

Figur 21

4.1.6 Alarm

Efter att ett alarm har utlöstes så går det att se vad det är som IDS:n har reagerat på i log filen. Det går att se logfilen med kommandot *tail /var/log/watfu/detection.log*. Alla detektioner har en tidstämpel, ett meddelande om vad som har märkts, protokoll och source med destinations adress. Efter att vi körde en skanning av 192.168.3.0/24 för att testa bas säkerheten. IPS:n gjorde inget åt alla dem ICMP paket men IDS:n märkte dock efter ett antal ping packet att det var något som pågick. BDsmIDS IDS:n kände av att det var en nätverksskanning med hjälp av att kolla source adressen och destination, då det var samma source adress till flera olika destination adresser under ett visst tidsspänn. (se figur 22)

```
12/07/2023-18:58:25.036474  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.13:0  
12/07/2023-18:58:25.036487  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.14:0  
12/07/2023-18:58:25.127542  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.17:0  
12/07/2023-18:58:25.127701  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.18:0  
12/07/2023-18:58:25.128081  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.20:0  
12/07/2023-18:58:25.128234  [**] [1:1:1] Nmap Scan [**] [Classification:  
[Priority: 3] {ICMP} 172.16.0.12:8 -> 192.168.3.21:0
```

Figur 22

5 Demonstration

https://www.youtube.com/watch?v=1QEdYIYfU_4&ab_channel=NetworkDuck

https://www.youtube.com/watch?v=jj4_CJNXuvo&ab_channel=NetworkDuck

6 Running-Configs

Allt är nedanför :)

6.1 ASA

: Saved

:

ASA Version 9.0(3)

!

hostname ASA

enable password 8Ry2YjIyt7RRXU24 encrypted

xlate per-session deny tcp any4 any4

xlate per-session deny tcp any4 any6

xlate per-session deny tcp any6 any4

xlate per-session deny tcp any6 any6

xlate per-session deny udp any4 any4 eq domain

xlate per-session deny udp any4 any6 eq domain

xlate per-session deny udp any6 any4 eq domain

xlate per-session deny udp any6 any6 eq domain

names

!

interface Ethernet0/0

shutdown

no nameif

no security-level

no ip address

!

interface Ethernet0/1

nameif OUTSIDE

security-level 0

ip address 172.16.1.2 255.255.255.0

!

interface Ethernet0/2

```
nameif inside
security-level 100
ip address 172.16.2.1 255.255.255.0
!
interface Ethernet0/3
shutdown
no nameif
no security-level
no ip address
!
interface Management0/0
shutdown
no nameif
no security-level
no ip address
!
ftp mode passive
dns domain-lookup OUTSIDE
dns server-group DefaultDNS
name-server 8.8.8.8
access-list DHCP extended permit udp 192.168.2.0 255.255.254.0 host 172.16.1.1 eq bootpc
access-list DHCP extended permit udp 192.168.2.0 255.255.254.0 host 172.16.1.1 eq bootps
access-list DHCP extended permit udp host 172.16.1.1 192.168.2.0 255.255.254.0 eq bootpc
access-list DHCP extended permit udp host 172.16.1.1 192.168.2.0 255.255.254.0 eq bootps
access-list DHCP extended permit tcp any 192.168.3.0 255.255.255.0 eq www
access-list NMAP extended permit icmp host 172.16.0.12 192.168.0.0 255.255.0.0
pager lines 24
mtu OUTSIDE 1500
mtu inside 1500
no failover
icmp unreachable rate-limit 1 burst-size 1
no asdm history enable
```



```
arp timeout 14400
no arp permit-nonconnected
access-group DHCP in interface OUTSIDE
!
router ospf 1
router-id 4.4.4.4
network 172.16.1.0 255.255.255.0 area 2
log-adj-changes
redistribute eigrp 666 metric 121 metric-type 1 subnets
!
router eigrp 666
eigrp router-id 4.4.4.4
network 172.16.2.0 255.255.255.0
redistribute static
!
route OUTSIDE 0.0.0.0 0.0.0.0 172.16.1.1 1
timeout xlate 3:00:00
timeout pat-xlate 0:00:30
timeout conn 1:00:00 half-closed 0:10:00 udp 0:02:00 icmp 0:00:02
timeout sunrpc 0:10:00 h323 0:05:00 h225 1:00:00 mgcp 0:05:00 mgcp-pat 0:05:00
timeout sip 0:30:00 sip_media 0:02:00 sip-invite 0:03:00 sip-disconnect 0:02:00
timeout sip-provisional-media 0:02:00 uauth 0:05:00 absolute
timeout tcp-proxy-reassembly 0:01:00
timeout floating-conn 0:00:00
dynamic-access-policy-record DfltAccessPolicy
user-identity default-domain LOCAL
http server enable
http 172.16.0.0 255.255.255.0 inside
http 192.168.3.0 255.255.255.0 inside
no snmp-server location
no snmp-server contact
snmp-server enable traps snmp authentication linkup linkdown coldstart warmstart
```

```
crypto ipsec security-association pmtu-aging infinite
crypto ca trustpool policy
telnet timeout 5
ssh timeout 5
console timeout 0
threat-detection basic-threat
threat-detection statistics access-list
no threat-detection statistics tcp-intercept
ntp server 172.16.1.1
ntp server 10.0.23.2
!
class-map global-class
  match any
class-map OUTSIDE-class
  match port tcp eq www
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
parameters
  message-length maximum client auto
  message-length maximum 512
policy-map global_policy
  class inspection_default
    inspect dns preset_dns_map
    inspect ftp
    inspect h323 h225
    inspect h323 ras
    inspect ip-options
    inspect netbios
    inspect rsh
```

```
inspect rtsp
inspect skinny
inspect esmtp
inspect sqlnet
inspect sunrpc
inspect tftp
inspect sip
inspect xdmcp
inspect icmp
policy-map HTTP
class OUTSIDE-class
ips inline fail-close sensor vs0
policy-map global-policy
class global-class
ips inline fail-close sensor vs0
!
service-policy global_policy global
service-policy HTTP interface OUTSIDE
prompt hostname context
no call-home reporting anonymous
call-home
profile CiscoTAC-1
no active
destination address http https://tools.cisco.com/its/service/oddce/services/DDCEService
destination address email callhome@cisco.com
destination transport-method http
subscribe-to-alert-group diagnostic
subscribe-to-alert-group environment
subscribe-to-alert-group inventory periodic monthly
subscribe-to-alert-group configuration periodic monthly
subscribe-to-alert-group telemetry periodic daily
Cryptochecksum:d41d8cd98f00b204e9800998ecf8427e
```

: end

6.2 Leaf-1

Current configuration : 12023 bytes

!

! Last configuration change at 17:09:18 UTC Fri Dec 8 2023

!

version 16.9

no service pad

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

! Call-home is enabled by Smart-Licensing.

service call-home

no platform punt-keepalive disable-kernel-core

!

hostname Leaf-1

!

!

vrf definition Mgmt-vrf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

enable secret 5 \$1\$k6xL\$v/FybSUIJSn2DHnj9Yxxk.

!

no aaa new-model

switch 1 provision ws-c3650-24ps

!

!

!

!

call-home

! If contact email address in call-home is configured as sch-smart-licensing@cisco.com

! the email address configured in Cisco Smart License Portal will be used as contact email address to send SCH notifications.

contact-email-addr sch-smart-licensing@cisco.com

profile "CiscoTAC-1"

active

destination transport-method http

no destination transport-method email

ip routing

!

!

!

!

!

no ip domain lookup

ip domain name pokaltagarna.se

!

!

!

no login on-success log

!

!

!

!

!

!

!

!

crypto pki trustpoint SLA-TrustPoint

enrollment pkcs12

revocation-check crl

!

crypto pki trustpoint TP-self-signed-3854951857

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3854951857

revocation-check none

rsakeypair TP-self-signed-3854951857

!

crypto pki trustpoint TP-self-signed-737098331

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-737098331

revocation-check none

rsakeypair TP-self-signed-737098331

!

crypto pki trustpoint TP-self-signed-3508284637

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3508284637

revocation-check none

rsakeypair TP-self-signed-3508284637

!

crypto pki trustpoint TP-self-signed-2661354057

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2661354057

revocation-check none

rsakeypair TP-self-signed-2661354057

!

!

crypto pki certificate chain SLA-TrustPoint

certificate ca 01

30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030

32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28

quit

crypto pki certificate chain TP-self-signed-3854951857

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 33383534 39353138 3537301E 170D3233 31323038 31333130
32335A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649

4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33 38353439
35313835 37308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 01009F36 B6BBCEF5 C707B8F2 8E883F65 C0C15A41 322D032E 47F52D2D
3055A1C4 4CD3E95D E58B3868 6187ADF0 612B6F77 331D6AA4 4B63FFD3 598AB901
30E2912A 37765355 A4E3D5E3 C0A72A19 95BCDFFE 8A5A5BEB 0CD8EFE1 ACA4A2E1
30944A48 EA7013E2 2203C296 680CB3DA 28982FCD B710275F 5B21D2C9 00BB1CDF
45B14745 4FD87914 94AE789F 1C14522D 0061851B 22D4CC25 5265BC76 3271CFE8
EE03E94F 3AEEE45F CACBCA20 D0AAFC2E 437366B9 8688F497 469824D8 478D898D
7BCB95D6 746CD088 D42979AB BC971F9D 763D9AA8 3B30D7BA 9ADE5B4D B607C847
E039A4AB 2169531C B943ADAC 7515EC7B 5F4A79AC ADD397A3 D6294794 9BC94A2C
13C2951D 1F7F0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 1418172E E7C38BD2 07560E96 2AD30FB8 E18D1FE7
D9301D06 03551D0E 04160414 18172EE7 C38BD207 560E962A D30FB8E1 8D1FE7D9
300D0609 2A864886 F70D0101 05050003 82010100 6CE6417D D3BEB49E 8F31A217
354F28B9 FC82CF08 7F2B0248 6AC25BDE 8B3100FC 250A60AA 8CB05165 90CA0ED3
ABBF39E7 4830FFF3 B8BDACB3 6389FB04 B3073BBC 5D414E5E A83B1973 AC8F046C
C17B40C9 BA9F0DA9 81E940E4 8B869FEB E19F675A 2767C13F 6AA0C26E 2314EF52
6289A52F 8D3AA71E 9E61540F 05D52ED9 92BBBAAD 19113DB7 877EED98 76EEE34C
00649514 96C5B61B AA663E8E 2618695B D50E876E 84F0E244 F3B1B86D 048F91E8
CA206836 F67E81D7 D4B7876C BC6F6431 257A3437 29886D62 1B7290C5 5ECA65A1
014D7E29 C7301284 E42A4407 21ACF170 D6C4CA61 79FB6624 657CD6A4 0B0B12E6
F3BA2800 38EC5154 53E9DBF9 D1545300 4D78A280

quit

crypto pki certificate chain TP-self-signed-737098331

crypto pki certificate chain TP-self-signed-3508284637

crypto pki certificate chain TP-self-signed-2661354057

!

license boot level ipservicesk9

!

!

diagnostic bootup level minimal

!

```
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 79468
!
!
username admin privilege 15 secret 9
$9$BtBZdhMQ5mL9N.$i9pSDUsjXOUxIbaWsJJDDw2NMWpdkpX6BdfxmOV9yH.
!
redundancy
mode sso
!
!
!
!
!
!
transceiver type all
monitoring
!
!
class-map match-any system-cpp-police-topology-control
  description Topology control
class-map match-any system-cpp-police-sw-forward
  description Sw forwarding, L2 LVX data, LOGGING
class-map match-any system-cpp-default
  description DHCP Snooping, EWLC control, EWCL data
class-map match-any system-cpp-police-sys-data
  description Learning cache ovfl, Crypto Control, Exception, EGR Exception, NFL SAMPLED DATA,
  RPF Failed
class-map match-any system-cpp-police-punt-webauth
  description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
  description L2 LVX control packets
class-map match-any system-cpp-police-forus
```

```
description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
class-map match-any system-cpp-police-multicast
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
class-map match-any system-cpp-police-system-critical
description System Critical and Gold
!
policy-map system-cpp-policy
!
!
!
!
!
!
!
!
```

!

!

!

interface GigabitEthernet0/0

vrf forwarding Mgmt-vrf

no ip address

shutdown

negotiation auto

!

interface GigabitEthernet1/0/1

description Link to Spine-1

no switchport

ip address 10.0.1.1 255.255.255.0

!

interface GigabitEthernet1/0/2

description Link to Spine-2

no switchport

ip address 10.0.2.1 255.255.255.0

!

interface GigabitEthernet1/0/3

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/4

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/5

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/6

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/7

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/8

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/9

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/10

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/11

description NOT IN USE

switchport access vlan 50

shutdown

!

```
interface GigabitEthernet1/0/12
description in use
no switchport
ip address 192.168.99.1 255.255.255.0
```

!

```
interface GigabitEthernet1/0/13
description NOT IN USE
switchport access vlan 50
shutdown
```

!

```
interface GigabitEthernet1/0/14
description NOT IN USE
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/15
description NOT IN USE
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/16
description NOT IN USE
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/17
description NOT IN USE
switchport access vlan 50
shutdown
```

!

interface GigabitEthernet1/0/18

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/19

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/20

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/21

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/22

description NOT IN USE

switchport access vlan 50

shutdown

!

interface GigabitEthernet1/0/23

description NOT IN USE

no switchport

ip address 192.168.69.1 255.255.255.0

!

interface GigabitEthernet1/0/24

```
description link to ASA
no switchport
ip address 172.16.2.2 255.255.255.0
ip summary-address eigrp 666 10.0.0.0 255.255.252.0
ip summary-address eigrp 666 192.168.2.0 255.255.254.0
!
interface GigabitEthernet1/1/1
description NOT IN USE
switchport access vlan 50
shutdown
!
interface GigabitEthernet1/1/2
description NOT IN USE
switchport access vlan 50
shutdown
!
interface GigabitEthernet1/1/3
description NOT IN USE
switchport access vlan 50
shutdown
!
interface GigabitEthernet1/1/4
description NOT IN USE
switchport access vlan 50
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan50
no ip address
```

```
shutdown
!
!
router eigrp 666
network 10.0.1.0 0.0.0.255
network 10.0.2.0 0.0.0.255
network 172.16.2.0 0.0.0.255
network 192.168.69.0
network 192.168.99.0
eigrp router-id 1.1.1.1
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip ssh version 2
!
!
!
!
!
!
control-plane
service-policy input system-cpp-policy
!
banner motd ^CCC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
!
line con 0
exec-timeout 0 0
password 7 121A0C041104
logging synchronous
login
```

```
stopbits 1
line aux 0
stopbits 1
line vty 0 4
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
line vty 5 15
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
!
!
monitor session 1 source interface Gi1/0/24
monitor session 1 destination interface Gi1/0/11 - 12
ntp server time.google.com
!
!
!
!
!
!
end
```

6.3 Leaf-2

Current configuration : 12295 bytes

```
!
! Last configuration change at 15:09:38 UTC Fri Dec 8 2023
!
version 16.9
no service pad
```

```
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
! Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
!
hostname Leaf-2
!
!
vrf definition Mgmt-vrf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
enable secret 9 $9$YiP18JTGAAarIE$4uwpD9kd.eYuWOugdS/pHmYyesizZ27XNJGjl2GpHYM
!
no aaa new-model
switch 1 provision ws-c3650-24ps
!
!
!
!
call-home
! If contact email address in call-home is configured as sch-smart-licensing@cisco.com
! the email address configured in Cisco Smart License Portal will be used as contact email
address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
```


active

destination transport-method http

no destination transport-method email

ip routing

!

!

!

!

!

no ip domain lookup

ip domain name pokaltagarna.se

!

!

!

login on-success log

!

!

!

!

!

!

!

!

crypto pki trustpoint SLA-TrustPoint

enrollment pkcs12

revocation-check crl

!

crypto pki trustpoint TP-self-signed-1857120401

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1857120401

revocation-check none

rsakeypair TP-self-signed-1857120401

!

crypto pki trustpoint TP-self-signed-3508284637

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3508284637

revocation-check none

rsakeypair TP-self-signed-3508284637

!

crypto pki trustpoint TP-self-signed-2661354057

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2661354057

revocation-check none

rsakeypair TP-self-signed-2661354057

!

crypto pki trustpoint TP-self-signed-737098331

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-737098331

revocation-check none

rsakeypair TP-self-signed-737098331

!

!

crypto pki certificate chain SLA-TrustPoint

certificate ca 01

30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101 0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317 43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533 30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355 040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73 696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1 F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388 8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7 D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F EA2956AC

7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF 58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B 42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8 8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368 95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04 04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604 1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01 010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78 240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1 6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146 8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2 55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49 1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69 39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD 230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F 719BB2F0
D697DF7F 28

quit

crypto pki certificate chain TP-self-signed-1857120401

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31383537 31323034 3031301E 170D3233 31323038 31333538
32325A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 38353731
32303430 31308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 01009523 C370B79A 433EC2C8 C0565A34 D2BA1C63 F18B32C3 DB731A4D
59A35C08 0807ACEF C315EBD3 C81DD58D 8DAA95C3 057D7357 89E91EFF 425AE71C
32CC6372 100F8260 DC20D7FA A87C20E7 34A8490B 8D1CDA42 036C825F 8AF38F41
048F7F81 9DB5BF6B 60688007 CE6EDBFC 3619FDA9 AFD249C5 95FFE16F 2B60C0B4
8EDDBE60 237A07D6 C6629845 28163CBE 78E5E513 C7D64114 5DBA14BD 2707BF0B
0508089C AD11607C 264AE59E 44B29523 D53CD5F7 F9981F09 729D2706 DBEFB255
72D87F61 32D3E9A5 F0A9BE5C B8F11BDB B36C8E64 F878FF09 77F832F2 0C7FAC0F

68E95427 1EA47DC6 E980AF38 08F2554E 5DFD82B0 D2E845F5 642B3250 A9F89F7F
F3243C90 088B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 145945F5 6C80F6BF BC8165B3 1ACBC872 E61AF703
4C301D06 03551D0E 04160414 5945F56C 80F6BFBC 8165B31A CBC872E6 1AF7034C
300D0609 2A864886 F70D0101 05050003 82010100 809F3787 D1C291AC 986C52FF
879FE5EB 90B29743 A67E20EC E12C0FF9 6ED080BB 81B41E54 AFB066DA E50179ED
07836053 63202192 D2CD907D 33751B3F 061FA5BA 1B3D7BA2 FA1613C5 404D1129
AFB47EBA A1840022 E721C8D8 2995984D D55673A5 C822BED0 530E87D7 DB920D70
4A302E08 724C5DE0 45268DB5 7C8CA86C EAFCCBB1A 450E9344 5F320AA0 3B0E731B
FB7F3E07 21AF58EB 230A6746 8FD4FAF7 8F090F8D 33F9FA4F 02F3A110 E642BE1B
4D6B6618 A4C1873E CDBAECB3 985E6B08 C5A7F434 C84055F5 D50C507B 076738A1
829E65E0 01CDAAA5 72A5F6EA 8522EFED B322E91E 7CF770DE 6572412A 7EEBF10C
8226E80A 9166013B 6E39421A AEAFF0EB3 7EF678AD

quit

crypto pki certificate chain TP-self-signed-3508284637

crypto pki certificate chain TP-self-signed-2661354057

crypto pki certificate chain TP-self-signed-737098331

!

license boot level ipservicesk9

!

!

diagnostic bootup level minimal

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

memory free low-watermark processor 79468

!

!

username admin privilege 15 secret 9

\$9\$62m0sbmG3AbDAU\$Mv13TCrkZyUQXDEevRHgPGy8AmEhjFwha82NyVeoSYk

!

redundancy

```
mode sso
!
!
!
!
!
transceiver type all
monitoring
!
!
class-map match-any system-cpp-police-topology-control
  description Topology control
class-map match-any system-cpp-police-sw-forward
  description Sw forwarding, L2 LVX data, LOGGING
class-map match-any system-cpp-default
  description EWLC control, EWLC data, Inter FED
class-map match-any system-cpp-police-sys-data
  description Learning cache ovfl, High Rate App, Exception, EGR Exception, NFL SAMPLED DATA,
  RPF Failed
class-map match-any system-cpp-police-punt-webauth
  description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
  description L2 LVX control packets
class-map match-any system-cpp-police-forus
  description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
  description MCAST END STATION
class-map match-any system-cpp-police-multicast
  description Transit Traffic and MCAST Data
class-map match-any system-cpp-police-l2-control
  description L2 control
class-map match-any system-cpp-police-dot1x-auth
```

```
    description DOT1X Auth
class-map match-any system-cpp-police-data
    description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
    description Stackwise Virtual
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
    description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
    description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
    description DHCP snooping
class-map match-any system-cpp-police-system-critical
    description UNUSED
!
policy-map system-cpp-policy
!
!
!
!
!
!
!
!
!
!
!
interface GigabitEthernet0/0
    vrf forwarding Mgmt-vrf
    no ip address
    shutdown
    negotiation auto
```

!

```
interface GigabitEthernet1/0/1
description link to SPINE-1
no switchport
ip address 10.0.1.2 255.255.255.0
```

!

```
interface GigabitEthernet1/0/2
description link to SPINE-2
no switchport
ip address 10.0.2.2 255.255.255.0
```

!

```
interface GigabitEthernet1/0/3
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/4
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/5
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
```

!

```
interface GigabitEthernet1/0/6
description UNUSED
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/7
```

```
description UNUSED
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/8
```

```
description UNUSED
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/9
```

```
description UNUSED
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/10
```

```
description UNUSED
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/11
```

```
description UNUSED
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```



```
!  
interface GigabitEthernet1/0/12  
description UNUSED  
switchport access vlan 50  
switchport mode access  
shutdown
```

```
!  
interface GigabitEthernet1/0/13  
description UNUSED  
switchport access vlan 50  
switchport mode access  
shutdown
```

```
!  
interface GigabitEthernet1/0/14  
description UNUSED  
switchport access vlan 50  
switchport mode access  
shutdown
```

```
!  
interface GigabitEthernet1/0/15  
description UNUSED  
switchport access vlan 50  
switchport mode access  
shutdown
```

```
!  
interface GigabitEthernet1/0/16  
description UNUSED  
switchport access vlan 50  
switchport mode access  
shutdown
```

```
!  
interface GigabitEthernet1/0/17
```

```
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/18
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/19
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/20
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/21
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/22
description UNUSED
switchport access vlan 50
```

```
switchport mode access
shutdown
!
interface GigabitEthernet1/0/23
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/24
description UNUSED
no switchport
ip address 192.168.2.1 255.255.255.0
ip helper-address 172.16.1.1
!
interface GigabitEthernet1/1/1
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/1/2
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/1/3
description UNUSED
switchport access vlan 50
switchport mode access
shutdown
```

```
!  
interface GigabitEthernet1/1/4  
  description UNUSED  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan50  
  no ip address  
  shutdown  
!  
!  
router eigrp 666  
  network 10.0.1.0 0.0.0.255  
  network 10.0.2.0 0.0.0.255  
  network 192.168.2.0  
  eigrp router-id 2.2.2.2  
!  
ip forward-protocol nd  
ip http server  
ip http authentication local  
ip http secure-server  
ip ssh version 2  
!  
!  
!  
!  
!
```

!

control-plane

service-policy input system-cpp-policy

!

banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C

!

line con 0

exec-timeout 0 0

password 7 104D000A0618

logging synchronous

login

stopbits 1

line aux 0

stopbits 1

line vty 0 4

exec-timeout 0 0

logging synchronous

login local

transport input telnet ssh

line vty 5 15

exec-timeout 0 0

logging synchronous

login local

transport input telnet ssh

!

!

monitor session 1 source interface Gi1/0/24

monitor session 1 destination interface Gi1/0/11 - 12

ntp server time.google.com

!

!

!

!
!
!
end

6.4 Leaf-3
Current configuration : 5749 bytes
!
!
!
version 15.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname Leaf-3
!
boot-start-marker
boot-end-marker
!
enable secret 9 \$9\$0yXk3Gk1T89q76\$7hvp7KUpX8TCzWlU.yNL8bt5g0zNnkaEq.4fL4yivE2
!
username admin privilege 15 secret 9
\$9\$.V2zWpikOPfAv.\$ZRZWCtqvMGyujLnE.3Tc4tgI2wyXpJuLza9wUsrcYsw
no aaa new-model
switch 1 provision ws-c2960x-24ts-l
!
!
!
!
ip routing
!
!

no ip domain-lookup

ip domain-name pokaltagarna.se

!

!

!

!

!

!

!

crypto pki trustpoint TP-self-signed-3022580736

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-3022580736

revocation-check none

rsakeypair TP-self-signed-3022580736

!

!

crypto pki certificate chain TP-self-signed-3022580736

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 33303232 35383037 3336301E 170D3233 31313238 31303034
31375A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D33 30323235
38303733 3630819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
8100A3B1 ABC9773A FDA1BD8A C726B3D7 E60540E3 EE9F8C78 687D8847 974DB33E
04E1E466 A448AD06 DB068A83 6CF02219 81B06E7A EF036A30 3F56AE29 78FC5AEE
A0A98C6E 0AED0BD6 800C91AF D1605D38 B0715700 21FC8A2F 2717D63A C4558305
1BB4F11A AEE8B816 7D3EE663 9E2DD2AF 1E4E12A4 F004420C 3BF037BF A530D87A
3A090203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 14C6A5F1 6E9A3377 0B814090 A719B058 973812BE 07301D06
03551D0E 04160414 C6A5F16E 9A33770B 814090A7 19B05897 3812BE07 300D0609
2A864886 F70D0101 05050003 8181008E 9865E6C2 FAC1A91C 435E0DF5 95921D7E

5C8379AD 887C75C3 B2204995 6F99FD5F 485111DE 26DBE5FE 9C7FE851 9EC69BFF
09B3CAE8 390BB508 6FC9E787 A246CEE3 362F4A44 03AC365A F7AD8B6D B9C1E4C3
49DF2A63 72EB4293 F4A3486E BD55DBF5 6F9C0541 113AB950 17713F3B F0AA6073
51D1AF8A 96421C38 57E4D6E2 6BC9A2

quit

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

!

!

!

vlan internal allocation policy ascending

!

!

!

!

!

!

!

!

!

!

!

interface FastEthernet0

no ip address

no ip route-cache

shutdown

!

interface GigabitEthernet1/0/1

description link to SPINE-1

no switchport


```
ip address 10.0.1.3 255.255.255.0
```

```
!
```

```
interface GigabitEthernet1/0/2
```

```
description link to SPINE-2
```

```
no switchport
```

```
ip address 10.0.2.3 255.255.255.0
```

```
!
```

```
interface GigabitEthernet1/0/3
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/4
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/5
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/6
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/7
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/8
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/9
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/10
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/11
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/12
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/13
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/14
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/15
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/16
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/17
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/18
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/19
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/20
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

!

interface GigabitEthernet1/0/21

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/22

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/23

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/24

no switchport

ip address 192.168.3.1 255.255.255.0

ip helper-address 172.16.1.1

!

interface GigabitEthernet1/0/25

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/26

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/27

```
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/28
switchport access vlan 50
switchport mode access
shutdown
!
interface Vlan1
no ip address
shutdown
!
interface Vlan50
no ip address
shutdown
!
!
router eigrp 666
network 10.0.1.0 0.0.0.255
network 10.0.2.0 0.0.0.255
network 192.168.3.0
eigrp router-id 3.3.3.3
eigrp stub connected summary
!
!
ip http server
ip http secure-server
ip ssh version 2
!
!
!
```

```
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
!
line con 0
exec-timeout 0 0
password 7 121A0C041104
logging synchronous
login
line vty 0 4
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
line vty 5 15
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
!
!
end
```

6.5 R1

Current configuration : 2410 bytes

```
!
! Last configuration change at 17:04:12 UTC Fri Dec 8 2023
!
version 16.6
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core
```

```

!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
 address-family ipv4
 exit-address-family
!
 address-family ipv6
 exit-address-family
!
enable secret 5 $1$pk4Q$ZTZ6YPXkb2UbRfJ9Kg8h51
!
no aaa new-model
!
no ip domain lookup
ip domain name pokaltagarna.se
!
!
!
!
!
!
!
!
!
!
!
subscriber templating

```

!

!

!

!

vtp domain VTP

vtp mode off

!

!

!

multilink bundle-name authenticated

!

!

!

!

!

!

license udi pid ISR4321/K9 sn FD024380JV3

diagnostic bootup level minimal

spanning-tree extend system-id

!

!

!

username admin privilege 15 secret 9

\$9\$ddaeCyxrZ/l/iR\$9NyoQFbUJQFbRc5YvWuBYv.tcTk5jywXAFVRuU8n1as

!

redundancy

mode none

!

!

!

!

!

!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!

```
interface GigabitEthernet0/0/0
ip address 10.0.13.1 255.255.255.252
negotiation auto
```

!

```
interface GigabitEthernet0/0/1
ip address 172.16.0.1 255.255.255.0
ip helper-address 10.0.13.2
negotiation auto
```

!

```
interface Serial0/1/0
ip address 10.0.12.1 255.255.255.252
```

!

```
interface Serial0/1/1
no ip address
```

```
shutdown
!
interface Serial0/2/0
no ip address
shutdown
!
interface Serial0/2/1
no ip address
shutdown
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 1.1.1.1
auto-cost reference-bandwidth 1000
passive-interface GigabitEthernet0/0/1
network 10.0.12.0 0.0.0.255 area 0
network 10.0.13.0 0.0.0.255 area 0
network 172.16.0.0 0.0.0.255 area 1
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ip ssh version 2
!
```

```
!  
!  
!  
!  
control-plane  
!  
banner motd ^CCC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C  
!  
line con 0  
  exec-timeout 0 0  
  password 7 0822455D0A16  
  logging synchronous  
  login  
  transport input none  
  stopbits 1  
line aux 0  
  stopbits 1  
line vty 0 4  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh  
line vty 5 15  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh  
!  
ntp server 10.0.12.1  
wsma agent exec  
!  
wsma agent config
```

!
wsma agent filesys

!
wsma agent notify

!
!
end

6.6 R2

Current configuration : 4849 bytes

!
! Last configuration change at 17:06:37 UTC Tue May 4 2021

!
version 16.6
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
platform qfp utilization monitor load 80
no platform punt-keepalive disable-kernel-core

!
hostname R2

!
boot-start-marker
boot-end-marker

!
!
vrf definition Mgmt-intf

!
address-family ipv4
exit-address-family

!
address-family ipv6
exit-address-family

!
enable secret 5 \$1\$ftsn\$swqCQOKrv/z8bao/CGtLU0

!
no aaa new-model
clock timezone UTC 1 0

!
no ip domain lookup
ip domain name pokaltagarna.se

!
!
!
!
!
!
!
!
!
!
!

subscriber templating

!
!
!
!

vtp domain cisco
vtp mode off

!
!
!

multilink bundle-name authenticated

!
!
!

crypto pki trustpoint TP-self-signed-2345577025

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-2345577025

revocation-check none

rsakeypair TP-self-signed-2345577025

!

!

crypto pki certificate chain TP-self-signed-2345577025

certificate self-signed 01

30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32333435 35373730 3235301E 170D3133 30313137 31343334
32355A17 0D323030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D32 33343535
37373032 35308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201 0100C27D 1F27E541 D8F55D74 0939BAA6 A9B1EF16 00D90495 6CBCB4E0
2CC88D70 CFA808BE 499A44FE 4969D04A A9FD7C34 AA40A607 CFA59227 92E4D977
E68B76C2 856A2651 96551C9A 2B344C35 84382DCB 17D19848 79AD4448 52AAEB69
1EDC4362 9E04308C 35AE5F23 FC810611 C612E9CC 7876C518 E499127A F3F422D5
CCFDA393 5AF930A2 9DA4EB66 14EAB1E2 832A9EF0 418F07B5 27FBF4A0 0CAB739D
16AF0248 C01541F0 20A59C52 A0562C7F 88707582 BF249BD5 FA63EC24 05038EDD
E5F69972 1FD88797 D6F6B296 56CDEEB0 94AFFC96 AB4E2646 27263570 6B0F51FD
16F32843 39A50A11 B114077A AC97F985 C26B054A 0A0E6046 FD5E2EA6 0CE0EA5E
5B147724 5F690203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603 551D2304 18301680 143753AC 8F2EBC14 A9D70414 AEF6C904 9BC3B4A3
2F301D06 03551D0E 04160414 3753AC8F 2EBC14A9 D70414AE F6C9049B C3B4A32F
300D0609 2A864886 F70D0101 05050003 82010100 47693BED BC9EBEB9 A74CC9EA
6154A963 07997B43 DDB571BA BC3075A9 E4515BCB 51D2393F 4249635B 4D01479B
75FA6761 32710F7D E9FF0F51 31D9AE54 244B5305 F2DD6B85 9B499A7E 977E6BD6
2A55B698 E22223F5 F81B7933 DB9DC67E FE2FABB1 32B9E050 2AA2203C CBAF50D6
0E7A4376 EE88C52B AC255916 BD7E7F12 88860F89 2BF76D19 3B4EBE53 F9F49341
E4E8D305 1E3CB04C 40BC68BA A3F362D6 88B8CB86 5336D99C DFA75119 71DAF57E

AA5ED2FF DE77B544 D7E270C1 D2392434 81CAA39E D6A6B699 522DA604 F7EC47D3
E269E2A9 8646E43B 2579EEB6 2F89531D 7DB5D844 39A91841 A2D072C2 87D74D8C
9A27CD18 C1470712 E0B9AD23 412B4FB5 E673E9F0

quit

!

!

license udi pid ISR4321/K9 sn FDO24380HZ1

diagnostic bootup level minimal

spanning-tree extend system-id

!

!

!

username admin privilege 15 secret 9

\$9\$Py4m4f7aeMYts7\$Fvi9LKgl.GXesTi8nZooULQB0SHMlsqqHtdqREhdd7k

!

redundancy

mode none

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!
!
!
!
!
!
!
!

interface GigabitEthernet0/0/0
description Link to INTERNET
ip address dhcp
ip nat outside
negotiation auto

!

interface GigabitEthernet0/0/1
ip address dhcp
ip nat outside
negotiation auto

!

interface Serial0/1/0
ip address 10.0.12.2 255.255.255.252
ip nat inside

!

interface Serial0/1/1
ip address 10.0.23.2 255.255.255.252
ip nat inside

!

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto


```
!  
router ospf 1  
  router-id 2.2.2.2  
  auto-cost reference-bandwidth 1000  
  passive-interface GigabitEthernet0/0/1  
  network 10.0.12.0 0.0.0.3 area 0  
  network 10.0.23.0 0.0.0.3 area 0  
  default-information originate metric-type 1  
!  
ip nat inside source list NAT interface GigabitEthernet0/0/1 overload  
ip forward-protocol nd  
no ip http server  
ip http authentication local  
no ip http secure-server  
ip http client source-interface GigabitEthernet0/0/1  
ip tftp source-interface GigabitEthernet0/0/1  
!  
ip ssh version 2  
!  
!  
ip access-list standard NAT  
  permit 192.168.0.0 0.0.255.255  
  permit 172.16.0.0 0.0.255.255  
  permit any  
!  
!  
!  
!  
control-plane  
!  
banner motd ^CCC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C  
!
```

```
line con 0
exec-timeout 0 0
password 7 13061E010803
logging synchronous
login
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
line vty 5 15
exec-timeout 0 0
logging synchronous
login local
transport input telnet ssh
!
ntp server 34.202.215.187
ntp server time.google.com
wsma agent exec
!
wsma agent config
!
wsma agent filesys
!
wsma agent notify
!
!
end
```

6.7 R3

Current configuration : 4836 bytes

!

! Last configuration change at 17:09:04 UTC Fri Dec 8 2023

!

version 16.6

service timestamps debug datetime msec

service timestamps log datetime msec

service password-encryption

platform qfp utilization monitor load 80

no platform punt-keepalive disable-kernel-core

!

hostname R3

!

boot-start-marker

boot-end-marker

!

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

enable secret 5 \$1\$IwnA\$Yx.55AY52EQywpGy905qW.

!

no aaa new-model

!

no ip domain lookup

ip domain name pokaltagarna.se

```
ip dhcp excluded-address 192.168.2.1 192.168.2.10
```

```
ip dhcp excluded-address 192.168.3.1 192.168.3.10
```

```
ip dhcp excluded-address 172.16.0.1 172.16.0.10
```

```
!
```

```
ip dhcp pool AREA_1
```

```
network 172.16.0.0 255.255.255.0
```

```
default-router 172.16.0.1
```

```
domain-name cisco.com
```

```
dns-server 8.8.8.8
```

```
!
```

```
ip dhcp pool LAN2
```

```
network 192.168.2.0 255.255.255.0
```

```
domain-name cisco.com
```

```
dns-server 8.8.8.8
```

```
default-router 192.168.2.1
```

```
!
```

```
ip dhcp pool LAN3
```

```
network 192.168.3.0 255.255.255.0
```

```
default-router 192.168.3.1
```

```
domain-name cisco.com
```

```
dns-server 8.8.8.8
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
!
```

```
subscriber templating
```

!

!

!

!

vtp mode off

!

!

!

multilink bundle-name authenticated

!

!

!

crypto pki trustpoint TP-self-signed-263984172

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-263984172

revocation-check none

rsa-keypair TP-self-signed-263984172

!

!

crypto pki certificate chain TP-self-signed-263984172

certificate self-signed 01

3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 32363339 38343137 32301E17 0D313330 31313731 34353732
385A170D 32303031 30313030 30303030 5A303031 2E302C06 03550403 1325494F
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3236 33393834
31373230 82012230 0D06092A 864886F7 0D010101 05000382 010F0030 82010A02
82010100 AC97A07F 083DC920 8E36E539 922F99E2 B43D51CD 5B0705D4 D5F61E33
E04C4502 8C362876 2F184D3E 33FCF58D B919D436 BF0B4422 8709BAB7 72443AC8
39C3A950 89CB529F FBBCA2D3 A9B91B9D 6939D500 3026DD81 99D9C26B 705408DF
8D5E0FD0 812315D9 CE33345D 94500AC5 7E9E9AA8 5EA43314 8F45DA46 91B66C1D
CE153EF3 7CDCCEB8 1F4BE68E D0FD9221 4697A09B 34634521 88BB6CBC 61607EBE

B0129A40 F34D2C8D A417B54E A5222CD5 215AD29A 97381DF0 51FBC520 0D6C3004
B297760E 8F03D3C3 3771442E 79F2E043 7C88A8A0 9B117071 06BF3627 48D06954
8FA55CBB F81D7FA6 9D3A677C BE31429E 1C9FE8D5 6AAF8EED C673711A 500F219B
D531B209 02030100 01A35330 51300F06 03551D13 0101FF04 05300301 01FF301F
0603551D 23041830 168014DE 4FA44DBC 22D2CC1C C220D984 19B913CA 1BACE930
1D060355 1D0E0416 0414DE4F A44DBC22 D2CC1CC2 20D98419 B913CA1B ACE9300D
06092A86 4886F70D 01010505 00038201 01009CD2 4EFC3C51 26A6D94A 679A3BD2
D71CA672 58786A43 1143B228 2631039F 3D6BD5B3 5A3252B4 A2A7602E 5BB174C1
E85EEB84 F451F14A 947047C1 CC4F1124 263D0B95 83C287C7 B26D9CF5 B40919A7
9B67AA5B C0459B7E C6820296 2D189F26 7C86987D BEBF9BA0 2F1FF2AC DD2A2327
7C18FB8B 5695FD8A 15F2D5B5 B8BC9802 D3791F2F 0C34B82D 45174A4F FCFDF239
48CC8D8F 6FB8E663 A24355A5 8F8D8655 9BDBF589 683B4B3B BCC7A17C 686568F7
9B38F7C0 69360F8F B2491AB9 5847AA1D B076F3F5 19FE541D F7472B74 2783B095
C17149FE F4CA8D65 2BE1C0DF 9CF1194B 6ACC2B44 21503854 EDF0CB02 3CCC3954
CE136F08 14C82934 D5E5CA44 98818E9D B9FC

quit

!

!

license udi pid ISR4321/K9 sn FDO24380HXL

diagnostic bootup level minimal

spanning-tree extend system-id

!

!

!

username admin privilege 15 secret 9

\$9\$taFqyjw/DvvzJM\$D/dyQGplJCKMbknjInodhIR/ATamhkSI38Mv0JgYS/Q

!

redundancy

mode none

!

!

!

!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!

```
interface GigabitEthernet0/0/0
ip address 10.0.13.2 255.255.255.252
negotiation auto
```

!

```
interface GigabitEthernet0/0/1
ip address 172.16.1.1 255.255.255.0
negotiation auto
```

!

```
interface Serial0/1/0
ip address 10.0.23.1 255.255.255.252
```

!

```
interface Serial0/1/1
```

```
no ip address
shutdown
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
router ospf 1
router-id 3.3.3.3
auto-cost reference-bandwidth 1000
network 10.0.13.0 0.0.0.255 area 0
network 10.0.23.0 0.0.0.255 area 0
network 172.16.1.0 0.0.0.255 area 2
!
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
!
ip ssh version 2
!
!
!
!
!
!
control-plane
!
banner motd ^CCC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C
!
```



```
line con 0
exec-timeout 0 0
password 7 1511021F0725
logging synchronous
login
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
exec-timeout 0 0
logging synchronous
login local
!
ntp server 219.239.35.0
wsma agent exec
!
wsma agent config
!
wsma agent filesys
!
wsma agent notify
!
!
end
```

6.8 Spine-1

Current configuration : 7595 bytes

```
!
! Last configuration change at 17:36:15 UTC Thu Dec 7 2023
!
version 15.2
no service pad
```

```
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname SPINE-1
!
boot-start-marker
boot-end-marker
!
enable secret 9 $9$0vkY6TqEKP4QC1$s9MheyBYVilmNK0jzEVXfXgjRtYiF9CA/iObPQTiER6
!
username admin privilege 15 secret 9
$9$SYdCeEkpUGiOf5$P6SfgEw4XPy6l6Iyq5xLoMxBLLyLcRfLKFNc7Y.Rc7U
no aaa new-model
switch 1 provision c1000-48t-4g-l
system mtu routing 1500
!
!
no ip domain-lookup
ip domain-name pokaltagarna.se
!
!
!
!
!
!
!
crypto pki trustpoint TP-self-signed-1931170048
  enrollment selfsigned
  subject-name cn=IOS-Self-Signed-Certificate-1931170048
  revocation-check none
  rsa-keypair TP-self-signed-1931170048
!
```

!

crypto pki certificate chain TP-self-signed-1931170048

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31393331 31373030 3438301E 170D3233 31323037 30393034
32385A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 39333131
37303034 3830819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
8100A94D 5DFCA35B 36394874 69D1779C 1CE048D9 6D203B9C 0BDB59C1 8EF444DB
C70C03B1 0446F363 080BF16B 460C2CF1 01C5E287 956FF5BF A5021450 2D54C9ED
A163CA17 788096F2 88147EF2 9D4954D3 2C420927 7F84022B B31C76E2 1A046AE4
B6AEA4C6 17EEE000 657CE0A4 00D49827 8405156A 15E3D613 EB03A270 450788B2
F3490203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 14B062C3 86965F06 1B06411E 1A2708C5 39EDE615 2C301D06
03551D0E 04160414 B062C386 965F061B 06411E1A 2708C539 EDE6152C 300D0609
2A864886 F70D0101 05050003 8181002C AA97F782 D3BEF8B9 06DB3A30 DA7E7FF8
3591159E F888FD8A 4C25D7A7 789F8812 D342980E 4DCFBE66 8ED985EA 17FECBAC
B87438D2 A461B149 12565CDE C70535D4 B9E2C0FF 0B88DD68 A6D5DF30 E7453ED7
A311D84F 4406B3CE 2AEBABA8 0C710478 BF08E535 37559B24 DF0450FE 6C839FED
D9ADE34A 2709AAB3 09D07E44 0417BA

quit

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

vlan internal allocation policy ascending

!

!

!

!

!

```
!  
interface GigabitEthernet1/0/1  
!  
interface GigabitEthernet1/0/2  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/5  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/6  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/7  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/8  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!
```

```
interface GigabitEthernet1/0/9
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/10
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/11
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/12
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/13
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/14
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/15
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/16
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/17
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/18
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/19
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/20
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/21
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

!

interface GigabitEthernet1/0/22

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/23

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/24

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/25

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/26

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/27

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/28

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/29
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/30
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/31
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/32
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/33
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/34
```

```
switchport access vlan 50
```

```
switchport mode access
```


shutdown

!

interface GigabitEthernet1/0/35

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/36

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/37

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/38

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/39

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/40

switchport access vlan 50

switchport mode access

shutdown

!

```
interface GigabitEthernet1/0/41
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/42
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/43
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/44
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/45
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/46
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/47
```

```
switchport access vlan 50
```

```
switchport mode access
shutdown
!
interface GigabitEthernet1/0/48
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/49
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/50
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/51
switchport access vlan 50
switchport mode access
shutdown
!
interface GigabitEthernet1/0/52
switchport access vlan 50
switchport mode access
shutdown
!
interface Vlan1
description Remote VLAN XD
ip address 10.0.1.254 255.255.255.0
shutdown
```

```
!  
interface Vlan50  
  no ip address  
  shutdown  
!  
ip http server  
ip http secure-server  
ip ssh version 2  
!  
!  
!  
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C  
!  
line con 0  
  exec-timeout 0 0  
  password 7 030752180500  
  logging synchronous  
  login  
line vty 0 4  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh  
line vty 5 15  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh  
!  
end
```

6.9 Spine-2

Current configuration : 7586 bytes

```
!  
! Last configuration change at 08:31:57 UTC Thu Dec 7 2023  
!  
version 15.2  
no service pad  
service timestamps debug datetime msec  
service timestamps log datetime msec  
service password-encryption  
!  
hostname SPINE-2  
!  
boot-start-marker  
boot-end-marker  
!  
enable secret 9  
$9$4OswE3wO6MEQcH$qSNvmvW84dnQkkSZ15N2PEFL55rqu9RkMrIpHQLD9qM  
!  
username admin privilege 15 secret 9  
$9$tldnwg/BDmWqRb$gTY/nDibZde.05S0P1.2GBz/uX5VlBJTg17QuxGO3Bo  
no aaa new-model  
switch 1 provision c1000-48t-4g-l  
system mtu routing 1500  
!  
!  
no ip domain-lookup  
ip domain-name pokaltagarna.se  
!  
!  
!  
!  
!  
!  
crypto pki trustpoint TP-self-signed-1931144576
```

enrollment selfsigned

subject-name cn=IOS-Self-Signed-Certificate-1931144576

revocation-check none

rsakeypair TP-self-signed-1931144576

!

!

crypto pki certificate chain TP-self-signed-1931144576

certificate self-signed 01

3082022B 30820194 A0030201 02020101 300D0609 2A864886 F70D0101 05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D 43657274
69666963 6174652D 31393331 31343435 3736301E 170D3233 31323037 30383331
35335A17 0D333030 31303130 30303030 305A3031 312F302D 06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31 39333131
34343537 3630819F 300D0609 2A864886 F70D0101 01050003 818D0030 81890281
81009197 6DC52471 3659B2F4 6DED4ABF F6339404 6927EFB0 FA04107D C62A5123
0C66C5A1 C51AB02C 364EE169 A11A9048 75875170 9EA653A2 4F2018B8 4A00FDB7
A6DDDF1B 5A934E41 A559AA74 4F888C7D 7369718D E68C830A 872B4190 9F075CCF
F7CAA647 C5F8FF37 47A03EB1 A1B79A48 49012BEC F61C6593 855044C2 481987EA
639B0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF 301F0603
551D2304 18301680 14087959 E7AA554E D2EEE13A 95A7BF8F 9380BC79 ED301D06
03551D0E 04160414 087959E7 AA554ED2 EEE13A95 A7BF8F93 80BC79ED 300D0609
2A864886 F70D0101 05050003 81810070 5F08E8D6 F1AFE105 AE164B6E C1B5B3DC
C8698B2D 9D151495 1FC9FD66 2E56E76A 987E533D FC706A40 E4117EB2 B4A1A1AD
E62B1CBF DF96B5E9 77EDB9F6 A8A8CFDE 11DA4688 65D009E0 CCA5FA6D 601B2795
A395FBF5 F0956241 AD96BF9D 607BE701 36EE8D09 8E6D23B8 AA6B8BB7 29A6BFA5
E2ED552E AB99D1E4 C6C13E30 3925BE

quit

!

spanning-tree mode rapid-pvst

spanning-tree extend system-id

!

vlan internal allocation policy ascending

```
!  
!  
!  
!  
!  
!  
interface GigabitEthernet1/0/1  
!  
interface GigabitEthernet1/0/2  
!  
interface GigabitEthernet1/0/3  
!  
interface GigabitEthernet1/0/4  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/5  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/6  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!  
interface GigabitEthernet1/0/7  
  switchport access vlan 50  
  switchport mode access  
  shutdown  
!
```

```
interface GigabitEthernet1/0/8
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/9
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/10
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/11
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/12
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/13
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/14
```

```
switchport access vlan 50
```



```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/15
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/16
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/17
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/18
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/19
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/20
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

!

interface GigabitEthernet1/0/21

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/22

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/23

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/24

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/25

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/26

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/27

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/28
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/29
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/30
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/31
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/32
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/33
```

```
switchport access vlan 50
```

```
switchport mode access
```

shutdown

!

interface GigabitEthernet1/0/34

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/35

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/36

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/37

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/38

switchport access vlan 50

switchport mode access

shutdown

!

interface GigabitEthernet1/0/39

switchport access vlan 50

switchport mode access

shutdown

!

```
interface GigabitEthernet1/0/40
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/41
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/42
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/43
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/44
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/45
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/46
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/47
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/48
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/49
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/50
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/51
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet1/0/52
```

```
switchport access vlan 50
```

```
switchport mode access
```

```
shutdown
```

```
!  
interface Vlan1  
  description MANAGEMENT ONLY  
  ip address 10.0.2.254 255.255.255.0  
!  
interface Vlan50  
  no ip address  
  shutdown  
!  
ip http server  
ip http secure-server  
ip ssh version 2  
!  
!  
!  
banner motd ^CC We are pokaltagarna, we do not forget, we do not forgive. Expect us. ^C  
!  
line con 0  
  exec-timeout 0 0  
  password 7 1511021F0725  
  logging synchronous  
  login  
line vty 0 4  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh  
line vty 5 15  
  exec-timeout 0 0  
  logging synchronous  
  login local  
  transport input telnet ssh
```

!

end